TerraQuest (Touch Grass AI)

Complete Hackathon Project Masterplan & Execution Guide

Theme: "Touch Grass" | **Core AI:** Google Gemma (Open-Weight) | **Prizes Targeted:** $600+

1. Executive Summary & Core Concept

TerraQuest is an open-source, gamified outdoor exploration and environmental action web application. It turns the real world into an adventure RPG where players receive real-world quests to explore green spaces, pick up trash, and plant saplings.

Unlike conventional apps that trap users in front of screens, TerraQuest follows the **"Minimal Screen Time" philosophy**: users check the app for 30 seconds to get their quest, listen to hands-free voice audio on the walk, spend 1 to 2 hours in nature, and upload photos for automated open-AI verification.

2. How It Wins the "Touch Grass" Theme

- **Screen is the Shortest Step:** App interactions are strictly brief checkpoints (receive quest -> take photos -> earn rank). All actual engagement occurs outdoors.
- **Time-Framed Multipliers (2x Bonus):** Dynamic golden-hour windows (e.g., 45-minute countdowns) motivate players to leave their desks immediately.
- **Real-World Eco Impact:** Every session cleans public spaces or promotes urban gardening, creating tangible positive externalities in neighborhoods.

3. "Why Open Innovation Matters" (The Winning Pitch Essay)

The hackathon prompt specifically asks: *"Tell us why open innovation matters for what you built."* TerraQuest answers this with 3 core pillars:

1. **Strict Neighborhood Privacy:** Quest photos contain user faces, home surroundings, and precise GPS coordinates. Closed proprietary APIs scrape or log this telemetry. By using open-weight models (**Google Gemma / PaliGemma**), inference can run on-device or on private community servers with zero external tracking.
2. **Zero Per-Token API Costs:** Environmental initiatives and community volunteer groups have zero budget for recurring API bills. Open-weight models democratize this verification technology for schools and non-profits worldwide.
3. **Domain Specialization:** Open models can be adapted with custom prompts or fine-tuned for specialized ecology tasks like local flora identification and litter categorization without vendor lock-in.

4. Targeted Sponsor Prize Categories ($600+ Potential)

Category

Prize
How TerraQuest Implements It
Best Use of Gemma
$200 USD
Core Verification & Quest Engine: Uses Google's open-weight model (Gemma / PaliGemma) to visually verify Before/After cleanup photos and generate localized outdoor challenges.
Best Use of SerpApi
$100 USD
Live Neighborhood Discovery: Grounds the agent in real local geographic data by pulling nearby public parks, botanical gardens, and trailheads dynamically.
Best Use of ElevenLabs
$100 USD
Hands-Free Audio Companion: Generates realistic adventurer voice narration for every quest so the user can keep their phone in their pocket and enjoy nature hands-free.
Best Use of Render
$200 USD
Live Cloud Deployment: Hosts the frontend and backend service live on Render's web runtime for instant, accessible judge testing.

5. Technical Architecture & AI Verification Pipeline

1. Location & Place Fetcher: Uses Geolocation + SerpApi to find genuine parks and green spaces within a 2 km radius.
2. Voice Synthesis: ElevenLabs narrates quest backstory ("Ranger Alert: Central Trail needs clearing").
3. Anti-Cheat & Metadata Gate:

- **EXIF Extractor:** Reads hardware camera tags, GPS coordinates, and timestamp (rejects downloaded web images).
- **Perceptual Hash:** Computes image fingerprint to block duplicate submissions.

4. Open-Weight Gemma Vision Engine:

- **Photo 1 (Before):** Verifies presence of visible trash/debris in outdoor setting.
- **Photo 2 (After):** Verifies that the scene matches Photo 1, but trash is removed and clean.
- **Score Calculation:** Computes Eco-Karma score and bonus XP multiplier.

5. Dual Leaderboard Engine:

- *Explorer XP:* Distance walked and timed quests finished.
- *Eco-Karma:* Verified trash cleanups and planted saplings (Ward / City level).

6. Hackathon Demo Video Script (2-Minute Walkthrough)

1. **0:00 - 0:25 (The Problem & Concept):** "Welcome to TerraQuest. Screens keep us trapped indoors. We built a platform powered by Google Gemma open-weight AI that gets you outside touching grass."
2. **0:25 - 0:50 (Live Quest & Audio Companion):** Show the location detector finding a nearby park. Click 'Play Audio' to hear the

ElevenLabs voice narrate the quest hands-free.

3 3 **0:50 - 1:20 (The AI Verification in Action):** Upload a Before photo (messy park bench) and After photo (clean bench). Show Gemma verifying the scene and awarding +250 Eco-Karma points.

4 4 **1:20 - 1:45 (Leaderboard & Open-Source Pitch):** Show the neighborhood leaderboard. Explain why Gemma and open weights ensure privacy and zero costs for community cleanups.

5 5 **1:45 - 2:00 (Conclusion):** "TerraQuest: Where AI serves to get you off the screen and into nature."

7. Submission Requirements Checklist

- • [x] **Working Demo Prototype:** Complete web app with live quest board, audio player, AI verification simulator, and leaderboards.
- • [ ] **Live Hosting Link:** Deployable to Render or Vercel.
- • [ ] **GitHub Repository:** Clean, documented code with README explaining Gemma, SerpApi, and ElevenLabs integrations.
- • [ ] **Demo Video / Screencast:** 2-minute video following the script above.
- • [ ] **Written Story:** The "Why Open Innovation Matters" section copied directly into your submission post.